

For the use of a Registered Medical Practitioner only

PRESCRIBING INFORMATION

SOBRIUM TABLETS

(Zolpidem Tartrate Tablets 10 mg)

COMPOSITION

Each film coated tablets contains:
Zolpidem Tartrate BP 10 mg

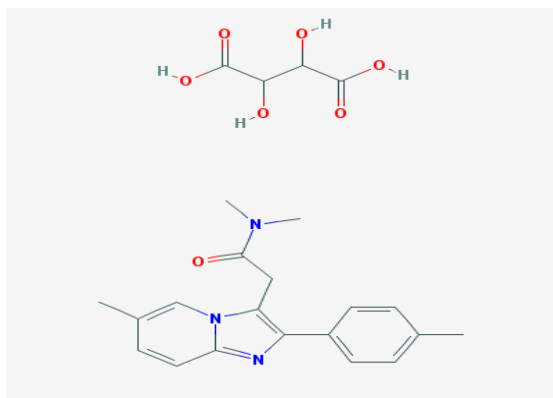
Excipients: Lactose monohydrate, microcrystalline cellulose, sodium starch glycollate, magnesium stearate, opadry OY 58910 white, purified water.

PRODUCT DESCRIPTION

White to off-white film coated, circular, biconvex tablets

DESCRIPTION

Zolpidem Tartrate is the tartrate salt of an imidazopyridine with non-benzodiazepine benzodiazepine-receptor agonist and sedative-hypnotic activities. It has a molecular formula of $C_{23}H_{27}N_3O_7$ and a molecular weight of 457.47638 g/mol. It has the following structural formula.



STRUCTURAL FORMULA
ZOLPIDEM TARTRATE

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- **Pharmacodynamics**

Pharmacologically, zolpidem binds selectively to the omega-1 subclass (or BZ1) of benzodiazepine receptors in the brain without binding to peripheral benzodiazepine receptors. This has been corroborated by the observation that zolpidem has little or no muscle relaxant properties.

Zolpidem has been shown to reduce sleep latency (time to fall asleep), decrease the number of awakenings, and to increase total sleep time. While REM sleep is not significantly decreased, the onset of REM is delayed. The REM/non-REM ratio is not significantly altered. Slow wave (stages 3 and 4) sleep time is increased. This more closely resembles natural sleep than does hypnosis induced by the benzodiazepines; this may result in fewer adverse reactions related to disturbance of normal sleep patterns.

- **Pharmacokinetics**

Absorption

Zolpidem is rapidly absorbed from the gastrointestinal tract after oral doses, peak plasma concentrations occurring within 3 hours. Oral bioavailability is 70%.

Effect of food: decreased systemic exposure (decreased C_{max} and AUC; increased T_{max}).

Distribution

Zolpidem is about 92% bound to plasma proteins. The distribution volume in adults is 0.54L/kg. Zolpidem is distributed into breast milk.

Metabolism

Metabolism is extensive in liver. Zolpidem undergoes first-pass metabolism. It is metabolised mainly by the cytochrome P450 isoenzyme CYP3A4.

Elimination

Zolpidem has an elimination half-life of about 2.5 hours. The inactive metabolites of zolpidem are excreted in the urine and faeces. Zolpidem is not dialysable.

Populations at risk

- In patients with renal insufficiency, whether dialysed or not, a moderate decrease in clearance is observed. Other kinetic parameters remain unchanged.
- In patients with hepatic insufficiency, the bioavailability of zolpidem is increased. Clearance is reduced and elimination half-life prolonged (about 10 hours).

INDICATIONS

SOBRIUM is indicated for the short-term treatment of insomnia. It should not be prescribed in quantities exceeding a 1-month supply.

DOSE AND METHOD OF ADMINISTRATION

The dose of **SOBRIUM** should be individualized.

The recommended dose for adults is 10 mg immediately before bedtime. Downward dosage adjustment may be necessary when zolpidem is administered with agents having known CNS-depressant effects because of the potentially additive effects.

Elderly or debilitated patients may be especially sensitive to the effects of zolpidem. Patients with hepatic insufficiency do not clear the drug as rapidly as normals. An initial 5-mg dose is recommended in these patients (see **PRECAUTIONS**).

The total zolpidem dose should not exceed 10 mg.

CONTRAINDICATIONS

Zolpidem tartrate is contraindicated in patients with a hypersensitivity to zolpidem tartrate or any of the inactive ingredients, obstructive sleep apnoea, myasthenia gravis, severe hepatic insufficiency, acute and/or severe respiratory depression. In the absence of data, zolpidem tartrate should not be prescribed for children or patients with psychotic illness.

WARNINGS AND PRECAUTIONS

Anaphylaxis (severe allergic reaction) and angioedema (severe facial swelling) which can occur as early as the first time the product is taken
Complex sleep – related behaviors which may include sleep driving, making phone calls, preparing and eating food while asleep

The cause of insomnia should be identified wherever possible and the underlying factors treated before a hypnotic is prescribed. The failure of insomnia to remit after a 7-14 day course of treatment may indicate the presence of a primary psychiatric or physical disorder, and the patient should be carefully re-evaluated at regular intervals.

Next-day psychomotor impairment

The risk of next-day psychomotor impairment, including impaired driving ability, is increased if:

- zolpidem tartrate is taken within less than 8 hours before performing activities that require mental alertness
- a dose higher than the recommended dose is taken;
- zolpidem tartrate is co-administered with other CNS depressants or with other drugs that increase the blood levels of zolpidem tartrate, or with alcohol or illicit drugs.

Zolpidem tartrate should be taken in a single intake immediately at bedtime and not be re-administered during the same night.

Respiratory Insufficiency:

As hypnotics have the capacity to depress respiratory drive, precautions should be observed if zolpidem is prescribed to patients with compromised respiratory function.

Use in patients with a history of drug or alcohol abuse:

Extreme caution should be exercised when prescribing for patients with a history of drug or alcohol abuse. These patients should be under careful surveillance when receiving zolpidem tartrate or any other hypnotic, since they are at risk of habituation and psychological dependence.

Psychotic illness:

Hypnotics such as zolpidem are not recommended for the primary treatment of psychotic illness.

Depression:

As with other sedative/hypnotic drugs, zolpidem tartrate should be administered with caution in patients exhibiting symptoms of depression. Suicidal tendencies may be present therefore the least amount of zolpidem that is feasible should be supplied to these patients to avoid the possibility of intentional overdose by the patient. Preexisting depression may be unmasked during use of zolpidem. Since insomnia may be a symptom of depression, the patient should be re-evaluated if insomnia persists.

General information relating to effects seen following administration of benzodiazepines and other hypnotic agents which should be taken into account by the prescribing physician are described below.

Tolerance:

Some loss of efficacy to the hypnotic effects of short-acting benzodiazepines and benzodiazepine like agents like zolpidem may develop after repeated use for a few weeks.

Dependence:

Use of benzodiazepines or benzodiazepine like agents like zolpidem may lead to the development of physical and psychological dependence. The risk of dependence increases with dose and duration of treatment; it is also greater in patients with a history of psychiatric disorders and/or alcohol or drug abuse.

These patients should be under careful surveillance when receiving hypnotics.

Once physical dependence has developed, abrupt termination of treatment will be accompanied by withdrawal symptoms. These may consist of headaches or muscle pain, extreme anxiety and tension, restlessness, confusion and irritability. In severe cases the following symptoms may occur: derealisation, depersonalisation, hyperacusis, numbness and tingling of the extremities, hypersensitivity to light, noise and physical contact, hallucinations or epileptic seizures.

Rebound insomnia:

A transient syndrome whereby the symptoms that led to treatment with a benzodiazepine or benzodiazepine like agent recur in an enhanced form may occur on withdrawal of hypnotic treatment. It may be accompanied by other reactions including mood changes, anxiety and restlessness.

It is important that the patient should be aware of the possibility of rebound phenomena, thereby minimising anxiety over such symptoms should they occur when the medicinal product is discontinued. Since the risk of withdrawal phenomena or rebound has been shown to be greater after abrupt discontinuation of treatment, it is recommended that the dosage is decreased gradually where clinically appropriate.

There are indications that, in the case of benzodiazepines and benzodiazepine like agents with a short duration of action, withdrawal phenomena can become manifest within the dosage interval, especially when the dosage is high.

Amnesia:

Benzodiazepines or benzodiazepine like agents such as zolpidem may induce anterograde amnesia. The condition occurs most often several hours after ingesting the product. In order to reduce the risk, patients should ensure that they will be able to have an uninterrupted sleep of 8 hours.

Other psychiatric and "paradoxical" reactions:

Other psychiatric and paradoxical reactions like restlessness, exacerbated insomnia, agitation, irritability, aggression, delusion, anger, nightmares, hallucinations, psychosis, abnormal behaviour and other adverse behavioural effects are known to occur when using benzodiazepines or benzodiazepine like agents. Should this occur, use of the product should be discontinued. These reactions are more likely to occur in the elderly.

Somnambulism and associated behaviours:

Sleep walking and other associated behaviours such as “sleep driving”, preparing and eating food, making phone calls or having sex, with amnesia for the event, have been reported in patients who had taken zolpidem and were not fully awake. The use of alcohol and other CNS depressants with zolpidem appears to increase the risk of such behaviours, as does the use of zolpidem at doses exceeding the maximum recommended dose. Discontinuation of zolpidem should be strongly considered for patients who report such behaviours (for example, sleep driving), due to the risk to the patient and others.

Severe injuries:

Due to its pharmacological properties, zolpidem can cause drowsiness and a decreased level of consciousness, which may lead to falls and consequently to severe injuries.

Children

Zolpidem should not be given to children aged under 18 years.

Effects on the ability to drive and use machines

Zolpidem has major influence on the ability to drive and use machines.

Vehicle drivers and machine operators should be warned that, as with other hypnotics, there may be a possible risk of drowsiness, prolonged reaction time, dizziness, sleepiness, blurred/double vision and reduced alertness and impaired driving the morning after therapy. In order to minimise this risk a resting period of at least 8 hours is recommended between taking zolpidem tartrate and driving, using machinery and working at heights.

Driving ability impairment and behaviours such as 'sleep-driving' have occurred with zolpidem tartrate alone at therapeutic doses.

Furthermore, the co-administration of zolpidem tartrate with alcohol and other CNS depressants increases the risk of such behaviours. Patients should be warned not to use alcohol or other psychoactive substances when taking zolpidem tartrate.

SOBRIUM TABLETS contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Concurrent use of rifampicin and zolpidem may result in decreased plasma concentration and pharmacodynamic effect of zolpidem.

Compounds which inhibit certain hepatic enzymes (particularly cytochrome P450) may enhance the activity of benzodiazepines and benzodiazepine-like agents.

Since CYP3A4 plays an important role in zolpidem tartrate metabolism, possible interactions with drugs that are substrates or inducers of CYP3A4 should be considered.

Concurrent use of deferasirox and CYP3A4 substrates may result in reduced plasma concentrations of CYP3A4 substrate.

Concurrent use of primidone and CYP3A4 substrates may result in decreased exposure of CYP3A4 substrates.

Concurrent use of zolpidem and ketoconazole may result in increased plasma concentrations and pharmacodynamic effects of zolpidem. Slight increase in sedation.

Concurrent use of ciprofloxacin and zolpidem may result in increased zolpidem plasma concentrations.

Concurrent use of fluvoxamine and zolpidem may result in decreased zolpidem clearance and increased exposure.

Concurrent use of zolpidem and ritonavir may result in an increased risk of extreme sedation and respiratory depression.

Concurrent use of zolpidem and the following may result in decreased zolpidem plasma concentrations:

- Carbamazepine
- St John's Wort

Concurrent use of zolpidem and the following drugs may result in an increased risk of hallucinations:

- Sertraline
- Venlafaxine
- Fluoxetine
- Bupropion
- Desipramine

In the case of narcotic analgesics enhancement of euphoria may also occur leading to an increase in psychological dependence.

Buprenorphine: Enhanced risk of respiratory depression that may prove fatal. Carefully assess the risk/benefit ratio of this combination.

Other Central nervous system depressants:

Morphine derivatives (analgesics, cough suppressants and replacement treatments, other than buprenorphine); neuroleptics; barbiturates; anxiolytic agents; other hypnotics; sedating antidepressants; antiepileptic drugs, anaesthetics, sedating H1 antihistamines; central anti-hypertensive drugs; baclofen; thalidomide; pizotifen.

Enhanced central nervous system (CNS)-depressant effects. Impaired vigilance may prove dangerous when driving or using machines. Dosage adjustments may be necessary when zolpidem is administered with sedative/hypnotic drugs because of the potentially additive effects.

Moreover, for morphine derivatives (analgesics, cough suppressants and replacement treatments), barbiturates: enhanced risk of respiratory depression which may prove fatal in the case of overdose.

Concurrent use of zolpidem and food may result in decreased zolpidem plasma concentrations. Zolpidem should not be administered with or immediately after a meal.

Concurrent use of zolpidem and alcohol may result in increased sedation.

UNDESIRABLE EFFECTS

Immune system disorders

Not known: angioneurotic oedema

Psychiatric disorders

Common: hallucination, agitation, nightmare

Uncommon: confusional state, irritability

Not known: restlessness, aggression, delusion, anger, psychosis, abnormal behaviour, somnambulism, dependence (withdrawal symptoms, or rebound effects may occur after treatment discontinuation), libido disorder, depression.

Most of these psychiatric undesirable effects are related to paradoxical reactions

Nervous system disorders

Common: somnolence, headache, dizziness, exacerbated insomnia, anterograde amnesia: (amnesic effects may be associated with inappropriate behaviour)

Not known: depressed level of consciousness

Eye disorders

Uncommon: diplopia

Respiratory, thoracic and mediastinal disorders

Not Known: respiratory depression

Gastro-intestinal disorders

Common: diarrhoea, nausea, vomiting, abdominal pain

Hepatobiliary disorders

Not known: Liver enzymes elevated

Skin and subcutaneous tissue disorders

Not known: rash, pruritus, urticaria, hyperhidrosis

Musculoskeletal and connective tissue disorders

Common: back pain

Not known: muscular weakness

Infections and infestations

Common: upper respiratory tract infection, lower respiratory tract infection

General disorders and administration site conditions

Common: fatigue

Not known: gait disturbance, drug tolerance, fall (predominantly in elderly patients and when zolpidem was not taken in accordance with prescribing recommendation)

Other adverse events reported with zolpidem include: constipation, indigestion, xerostomia, allergy, drugged state, asthenia, ataxia, cerebrovascular disease, difficulty driving a car, disorientated, lethargy, lightheadedness, motor retardation, sleep disorder, vertigo, hepatic encephalopathy, visual disturbance, abnormal vision, chest pain, tachycardia hypertension, palpitations, anaphylaxis, complex mannerisms – behavior, worsening, suicidal thoughts, anxiety, dream disorder, euphoria, memory impairment, abnormal liver function, arthralgia, myalgia, urinary tract infection, hiccoughs, pharyngitis, sinusitis, influenza-like illness.

Withdrawal sign or symptom: Withdrawal symptoms, including convulsions, tremor, abdominal and muscle cramps, vomiting, sweating, mild dysphoria, and insomnia have been reported with abrupt discontinuation of sedative/hypnotics. Other withdrawal effects reported include fatigue, nausea, flushing, lightheadedness, uncontrolled crying, emesis, panic attack, nervousness, and abdominal discomfort.

USE IN SPECIAL POPULATIONS

- **Pregnancy**

Currently insufficient clinical information is available concerning exposure during the first trimester of pregnancy.

By analogy with related drugs (benzodiazepines):

- If high-doses of zolpidem are being taken during the second and/or third trimesters of pregnancy, a reduction in active fetal movements and in fetal heart rate variability may occur.
- Treatment towards the end of pregnancy with benzodiazepines, even at a low dosage, may cause signs of impregnation in the neonate, such as axial hypotonia and difficulty in suckling which gives rise to poor weight gain. These signs are reversible, but may last for 1 to 3 weeks, depending on the half-life of the benzodiazepine prescribed. At high doses, reversible respiratory depression or apnea and hypothermia may appear in the neonate.

Furthermore, a neonatal withdrawal syndrome is possible, even if no signs of impregnation are present. This is characterized in particular by over-excitability, agitation and tremor in the neonate, occurring sometime after delivery. The time to onset depends on the elimination half-life of the drug and may be considerable if the latter is long. Hence it is preferable, as a precautionary measure, not to use zolpidem at any stage during pregnancy.

High doses of zolpidem should not be administered during the last trimester of pregnancy, as hypotonia and respiratory distress may occur in newborns at delivery. Withdrawal symptoms may occur in newborns a few days or weeks after birth.

- **Lactation**

Breast-feeding is not recommended during treatment.

OVERDOSE

Symptoms

Mild to moderate poisoning: Somnolence, slurred speech, confusion, and ataxia may occur.

Severe poisoning: Severe effects are very rare but may occur after co-ingestion with other sedatives and may include coma and respiratory depression. Death is extremely rare but may be caused by respiratory depression. Patients that present with coma are at risk for aspiration pneumonia, rhabdomyolysis, and renal failure.

Treatment

Majority of patients develop mild to moderate toxicity, and only require supportive care.

Severe toxicity generally occurs if other sedating agents are also ingested. Administer activated charcoal if the ingestion is recent and the patient is alert or the airway is protected. Orotracheal intubation for airway protection should be performed if the patient is increasingly drowsy or comatose.

STORAGE

Store below 30°C, protected from moisture.

KEEP MEDICINES OUT OF THE REACH OF CHILDREN

SUPPLY

Blister of 10 x 10's and 3 x 10's

Date of Revision: 21 September 2017

PRODUCT REGISTRATION HOLDER/ MANUFACTURED BY:

RANBAXY (MALAYSIA) SDN, BHD

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